

Quick tip: Review the prerequisites before you run the lab

Start Lab 01:30:00

Generative AI with Vertex AI: Text Prompt Design: Challenge Lab

Lab 1 hour 30 minutes No cost Introductory

★★★★☆

This lab may incorporate AI tools to support your learning.

Lab instructions and tasks —/100

Overview

Objective

Task 1. Open the notebook in Vertex AI Workbench

Task 2. Open the notebook and complete the challenge

Task 3. Progress checks

Congratulations!

Overview

This challenge lab is designed to test your knowledge of calling Gemini and utilizing a few fundamental text prompt design techniques.

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1. [Overview of prompting strategies](#) from the Generative AI on Vertex AI documentation.
2. [Prompt design strategies](#) from the Gemini API documentation.

Both contain good tips. You are encouraged to **bookmark them**.

Objective

You will demonstrate your ability to:

- Initialize Gen AI in your environment
- Load a generative model
- Guide model output with a persona
- Extract information to a schema
- Stay on topic with fallback responses
- Use examples to influence the model's response

and tasks that need to be completed, indicated by the code comments starting with # TODO. Your challenge is to complete each cell, run it to test for correctness, and then move on. When all the cells are working, you have completed the challenge.

Note: If you need help, [this notebook demonstrates getting started using Gemini in Python](#).

Task 1. Open the notebook in Vertex AI Workbench

1. In the Google Cloud Console, on the **Navigation menu**, click **Vertex AI > Workbench**. Alternatively, you can type "Vertex AI Workbench" in the top search bar and click the first result.
2. Find the `Workbench instance name` instance and click on the **Open JupyterLab** button.

tab.

Task 2. Open the notebook and complete the challenge

1. Click on the `notebook name` file.
2. In the **Select Kernel** dialog, choose **Python 3** from the list of available kernels.

- For **Project ID**, use **Project ID**, and for the **Location**, use **Region**.

3. Complete the challenge based on the notebook.

Note: Save your notebook before each progress check.

Task 3. Progress checks

After you run the notebook, check that you completed each task.

- Click **Check my progress** to verify the objective.

☐	Install & initialize Gen AI and a generative model Check my progress
☐	Use a persona to guide the model's response Check my progress
☐	Extract data from text to a structured format
☐	Use a fallback response to stay on topic Check my progress
☐	Use examples to guide model output Check my progress

Congratulations!

If you have completed the steps above, you have demonstrated your ability to use several prompt engineering techniques.

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